WELDING OF PRESSURE CONTAINING PIPING AND EQUIPMENT

FWEL CONTRACT No : 1-14-16601/19

CONTRACT NO. : POGC-760-81-156

PROJECT : SOUTH PARS GAS FIELD DEVELOPMENT (PHASES 9&10)

COMPANY : NATIONAL IRANIAN OIL COMPANY

SITE : ASSALUYEH / SOUTH PARS GAS FIELD PERSIAN GULF, IRAN

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GENERAL STATEMENT

The attached TOTAL EXPLORATION PRODUCTION (TEP) General Specification SP - TCS - 611 Rev. 6 is confirmed as the Project Specification, except as amended herein, and renumbered/reissued as specification N° RP-6340S-999-6300-001 Rev. 3 SOUTH PARS PHASES 9 & 10 Project.

3

ADDENDUM

The following items refer to the attached TEP General Specification
No. SP - TCS –611 *Rev. 6*.
The clauses set out below shall modify or replace the existing clauses in the original specification as noted.

3

GENERAL COMMENTS

- The following terms which are used in this general Job Specification are deleted and replaced by :
  - Engineering company by Contractor: shall mean the party which has entered under a Contract Agreement with POGC for the design, engineering, procurement and construction of the Onshore Facilities regarding the South Pars Field Development Project, Phases 9 & 10.
  - TOTAL: shall be read as POGC or his nominated Representative.

  - Manufacturer: shall mean any person, firm or company which manufacture or supply equipment or services for the performance of any item of the work.

  - Work: shall mean works and / or services entrusted by contractor under the Contract with company.
- Correct the following definition :
  - Inspection Agency: this term designates the Inspection Bodies which work under the supervision of POGC. In some cases, Contractor nominated by POGC may also act as the Inspection Agency.

1. SCOPE

Replace :

- SP-TCS-615 by : RP-6340S-999-6300-005 △3
- SP-TCS-620 by : RP-6340S-999-6300-006
- *Delete reference to "Duplex stainless steel (UNS 31803) : SP-TCS-614 ("Welding of duplex stainless steels")"*
- *Delete reference to Titanium (piping only) : SP-TCS-615 ("Welding of titanium pipes and fittings")*
- *Delete reference to "SP-TCS-617 ("Welding of 3 ½ % Ni Steel")*

Paragraph 3 Contractual reference documents

Replace :

- SPTCS 612 by : RP-6340S-999-6300-002 △3

Add :

RP-6340S-999-6300-003 "Metallurgical Requirements for Carbon Steel in Sour Wet Service" △3
(SSCC only).

Paragraph 3.2a) Replace:

ANSI B31.3 "Petroleum Refinery piping by ASME B31.3 "Process Piping".

Paragraph 3.2c)

Delete reference to A5-1, A5-5 and A5-9 and replace with "A5-X or A5-XX AWS Filler Metal Specifications". △3

Paragraph 4.1g)

Add the following:

Procedure qualification tests for CrMo steels (P number 3, 4 and 5) Replace 5Kg by 10Kg load.

***Paragraph 4.1h)**, second sentence*

"travel speel" should read "travel speed". △3

Paragraph 4.1k)

First paragraph

Delete "Specimens for chemical analysis shall be taken on the final weld layer" Replace with "Specimens for chemical analysis shall be taken at 3mm from surface of final layer" △2

Second paragraph

Add "When PWHT is required the Ferrite shall be in the range of 3 to 8 FN."

Paragraph 5.1.3 – Storage

Add new fourth paragraph as follows;

"Re-baking of electrodes is prohibited for those electrodes where it is prohibited by the manufacturers instructions e.g. vacuum packed electrodes" 3

Paragraph 6.1 a)

Add to:

"Double sided weld joints must have the root pass entirely removed by arc-air gouging, followed by grinding for final cleaning, prior to back welding".

The following sentence:

"or by grinding or machining".

Paragraph 6.1

Add new Paragraph 6.1g)

"For joint preparation in clad materials refer to 7.5.2." 3

Paragraph 7.2.1 Approved Welding Processes

(First paragraph) replaced by:

7.2.1.a) Pressure Vessels

For pressure retaining welds, following welding processes are acceptable:

Welding Processes	AWS Designation
Shielded Metal Arc Welding	SMAW
Gas Tungsten Arc Welding	GTAW
Automatic Submerged Arc Welding	SAW
Flux Cored Arc Welding	FCAW (1)

Note: 1. Flux Cored Arc Welding (FCAW) is restricted to carbon steel provided that the following conditions are fulfilled:

a. Application of FCAW process is limited to weld joints which cannot be welded with SAW automatic process (e.g. branch connection nozzles, horizontal, vertical welds ---).
b. Manufacturer must prove evidence of successful previous experience with this process.
c. The flux cored arc welding process is protected with external shielding gas.
d. In addition to the required non destructive examination, pressure retaining butt welds of thickness greater than 12.5mm shall be 100% examined by Ultrasonic Examination. For thickness equal or below 12.5mm pressure retaining welds not subject to radiography shall be examined by magnetic particle examination or dye penetrant examination.

- Other welding processes shall not be assumed acceptable by the Manufacturer during bid preparation and would require Engineering prior approval. Any authorisation would imply the evaluation of the specific application, additional qualifications and NDT requirements.

7.2.1.b) Piping

For Piping Pressure retaining welds, welds shall be limited to the following welding:

Welding Processes	AWS Designation
Shielded Metal Arc Welding	SMAW
Gas Tungsten Arc Welding	GTAW
Automatic Submerged Arc Welding	SAW
Flux Cored Arc Welding	FCAW (1)

Other welding processes shall not be assumed acceptable by the Manufacturer during bid preparation and would require Engineering prior approval. Any authorisation would imply the evaluation of the specific application, additional qualifications and NDT requirements.

Above limitations do not apply to non-pressure retaining welds.

Note: 1. Flux Cored Arc Welding (FCAW) may be used for piping prefabrication of carbon steel of rating up to 600lbs provided that the following conditions are fulfilled:

a. Manufacturer must prove evidence of successful previous experience with this process.
b. The flux cored arc welding process is protected with external shielding gas.
c. Welding is carried out at prefabrication using fully automatic machine equipment.

Flux cored arc welding without external shielding gas may only be used for welding non pressure containing parts and non pressure containing attachments to pressure containing parts.

Flux cored arc welding procedures shall be re-qualified whenever a change is made from one filler metal manufacturer to another or to a different brand or type from the same manufacturer.

Paragraph 7.2.1 Approved Welding Processes

(second paragraph) delete:

"or GMAW"

3

Replace Cr content "1.5%" by "above 2.9%".

Add new Paragraph 7.4 Site Welding

"For pressure vessels that are assembled on site, a production weld test plate is required for those seams produced at site. A production test plate is required for each welding process and welding position."

3

Re-number "Paragraphs 7.4 and 7.5" as "Paragraphs 7.5 and 7.6" 3

Paragraph 7.6 "Corrosion resistant overlays" 3

Add :

WELDING PROCESSES

The following processes are acceptable for application of overlay or clad restoration.

1. Gas Tungsten Arc Welding	GTAW
2. Submerged Arc Welding	SAW
3. Gas Metal Arc Welding	GMAW
4. Shielded Metal Arc Welding	SMAW
5. Electroslag Strip Weld Overlay	ESW
6. Flux Cored Arc Welding	FCAW

7. The use of other processes such as Plasma Arc Welding requires the prior authorisation of the Company.

Add :

For clad restoration of nickel or nickel-base alloys, the filler material shall match with the clad material composition.

Add :

Fabrication welding requirements

Minimum thickness of weld overlay after grinding and surface preparation shall be 3 mm unless otherwise specified in the contractor's design drawings or data sheets. Maximum overlay thickness shall not exceed 9 mm.

Multiple-layer overlay welds shall have the same number of layers as those qualified on the welding procedure specification.

The weld overlay shall be deposited in such a way that weld passes are orientated circumferentially around the inside of the vessel. In the case of smaller diameter nozzles less than 100 mm inside diameter, the weld passes may be deposited in the longitudinal direction.

Weld overlay surface shall be relatively smooth with no notches and undercuts that would act as stress intensifiers.

Add :

VISUAL EXAMINATION

Welds shall be examined visually before any other non-destructive examinations are performed.

All cracks, lack of fusion, surface slag /scale overlaps, undercuts, arc strikes and surface porosity are unacceptable.

Add :

LIQUID PENETRANT EXAMINATION

All weld deposit overlay including clad restoring at weld seams, whether by manual, or automatic procedure, shall be 100% liquid dye penetrant examined.

Weld deposit overlay machined surfaces shall be 100% PT on finished condition.

All cracks and circular defects greater than 1.6 mm diameter in weld deposit overlay shall be removed. Repaired areas shall be 100% re-inspected by liquid dye penetrant.

Paragraph 8. Preheating

Carbon Equivalent formula should read as;

3

$$CE = C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15$$

Paragraph 10 b) "Radiographic inspection"

First paragraph: replace "X rays shall be used exclusively unless otherwise agreed by TOTAL" by "unless otherwise specified by the Engineer, radiographic examination requirements shall be in accordance with the applicable code".

Delete the first sentence of the fourth paragraph and replace with;

" Sensitivity of films shall be verified by the use of wire type Image Quality Indicator (IQI) to ASTM E747 or hole type IQI to ASTM E142 placed transverse to the weld on the source side for each exposure (as an alternative IQI's in accordance with DIN standards may be used).

3

Paragraph 10d)

Delete "For spot radiographed equipment, this shall not be less than 10% of the weld length" and replace with "For spot radiography, a minimum of 10% of the weld length including circ/long seam T junctions shall be radiographed"

3

Add following New paragraph : **10 h)**

h) Qualification of NDT personnel:

– all operators, including for Dye Penetrant or Magnetic Particle Inspection, must be qualified to SNT-TC-1A level I, as a minimum or COFREND or CSWIP or EN473.

– interpretation must be carried out by inspectors holding at least a valid Certificate to SNT-TC-1A level II or CSWIP level II or COFREND level II or EN473 level II.

TABLE 4 Post Weld Heat Treatment for carbon steels and ferritic alloy materials

- *Delete "Minimum preheat temperature" and replace with "Minimum PWHT Temperature".*
- *Add new note 3 "3. Flange facings of vessel flanges (e.g. channel flanges) shall be machined after PWHT"* 3
- *Add new note 4 "4. NDE and hardness tests required shall be performed after completion of PWHT"*
- *Add new note 5 "5.Welding is not permitted on the pressure retaining shell after PWHT"*

APPENDIX 1
PD 5500 APPLICATION

Add :

The following items refer to attached TEP General Specification n° SP-TC-611 revision 6, amended and reissued as specification n° RP 6340 999 6300 01 Rev 2.

When PD 5500 code is applicable, the clauses set out below shall modify or replace existing clauses in the original specification as noted. 2

GENERAL COMMENTS

Paragraph 1 "SCOPE"

Cancel : "ASME and ANSI codes and standards" and replace by "PD 5500" 2

Paragraph 3.2 "Applicable codes and standards"

Delete and replace by :

The PD 5500 code.

The applicable issues are the latest revisions at date of call for bid. 2

Paragraph 4.1 a)

Add :

Welding procedures shall be qualified in accordance with section 5.2 of PD 5500. 2

Paragraph 4.2 "Welders and welding operators"

(First paragraph) Cancel : "ASME, section IX" and replace by "section 5.3 of the PD 5500". 2

Paragraph 5. "Welding consumables"

Replace :

In Table 1

ASME filler metal for welds joining similar materials by: Equivalent filler metal designated by the equivalent BS norms.

In Table 2

ASME filler metals for welds joining dissimilar materials by: Equivalent metal designated by BS norms.

Paragraph 8 - Table 3

Delete the table under the title and replace by :

See PD 5500, paragraph 4.4.1 "Preheat requirements"

Paragraph 9 a) "Post-weld heat treatment"

Cancel :

"applicable ASME or AINSI B31.3 code" and replace by: "PD 5500 code".

Paragraph 9 Table 4

Delete the table under the title and replace by :

See PD 5500, paragraphs 4.4.3 to 4.4.5.

Paragraph 10 a)

Cancel :

"ASME or ANSI B31.3 code and replace by: "PD 5500 code" 2

Paragraph 10 Table 5

Delete the table under the title and replace by :

See PD 5500, paragraphs 5.6 and 5.7.

Paragraph 10 b) "Radiographic inspection"

(Fourth paragraph) replace : 2

IQI of DIN standard by: Paragraph 5.6.6.1 of PD 5500

Add to the 6th sentence :

"The sensitivity maximum acceptable values, as a function of thickness examined, are given in the following table".

The following "When the value given in the PD 5500 code is more stringent than the value given in table 6, the PD 5500 code is applicable". 2

Paragraph 10 h) Qualification of NDT personnel:

Replace SNT-TC-1A by:

European Norm EN473 entitled: Qualification and Certification of NDT personnel and/or para 5.6.1. of PD 5500.

APPENDIX 2
WELDING OF TUBE - TUBE SHEET CONNECTIONS

1. INTRODUCTION

This chapter is applicable when strength welded tube/tube sheet connections have been specified and is intended to supplement EEMUA Publication N° 143 (formerly OCMA Specification n° TEW 1).

This chapter is applicable to shell and Tube Heat Exchangers and Air Cooled Heat Exchanger Equipment.

2. WELDING PROCEDURE QUALIFICATION

No production welding shall be carried out until the proposed welding procedures have been evaluated and authorised by the CONTRACTOR in accordance with the requirements of this chapter.

Welding procedure shall be certified by a recognised Inspecting Authority.

Procedure Test Piece

The procedure shall be welded in the same position as the actual welding to be done, i.e. vertical or downhand. Also, the welding sequence/configuration shall be part of the procedure qualification.
All tubes used for procedure testing shall be of the same diameter, wall thickness and nominal composition as those proposed for production. Plate material to be used for the procedure test shall also be of the same nominal composition as that to be used in manufacture. The thickness of the sample tube plates shall equal that of the plate to be used in production, except that it need not exceed 50mm thickness. If tube expansion after welding is specified, it may be necessary for a sample of full plate thickness to be employed. The test piece shall be at least 25mm greater in size all round than the limits of the tube hole array.

2.1 Examination of Test Pieces

In addition to the mandatory tests defined in EEMUA Publication n° 143, the following testing requirements shall apply:
Two of the tubes shall be cut for macroscopic examination in such a way that the starting point of the weld can be examined.
Hardness survey (HV*10* Vickers) shall be carried out across two macro examination specimens, for ferritic steels only. 3
Two sets of micrographic weld examination at a magnification of 200 shall be performed to show the weld, H.A.Z. such as to include tube and tube sheet material.
Two pull out tests for stationary tube sheets.

2.2 Weld Quality/Acceptance criteria

In addition to EE MUA publication n° 143 the following acceptance criteria shall apply:

- Hardness survey shall meet the requirements defined in paragraph 4 of this specification, except for sour service application, which shall be in accordance with RP 6340 999 6300 002 (SP - TCS - 612).
- For micrography interpretation, un-tempered martensitic structure shall not be acceptable;

3. WELDING PROCESS

3

- Only G.T.A.W. welding process shall be allowed
- Welding shall include two-weld passes minimum for strength welds unless the thickness of the tubes is less than 1.6 mm.
- The use of other processes require the prior acceptance of the Company.

4. EXAMINATION OF PRODUCTION WELDS

All complete welds shall be examined by dye penetrant. After dye penetrant testing and before any further welding is carried out, all traces of dye, developer and solvents shall be removed.

5. PRODUCTION MOCK-UP TEST

Unless otherwise specified on data requisition one mock-up production coupon shall be carried out concurrently with the fabrication of each individual heat exchanger of rating 1500 Lbs and above. Test requirements shall be in accordance with paragraph 2 of this Appendix 2.

6. REPAIRS

Faulty welds shall be completely removed to sound metal and repaired using qualified repair procedure.

Minor faults associated with fully automatic GTAW welding incorrect settings or tracking, may be rectified by re-welding at corrected settings or locally repaired by manual GTAW process. When any defects occur, particularly cracking, the cause shall be established prior to repair.

Any repair procedure shall be subject to repair procedure testing which shall consist of one macro-hardness test survey. Acceptance criteria shall be in accordance with paragraph 2.2 of this Appendix.

7. LEAK DETECTION

Leak detection shall be in accordance with RP 6340 999 0600 001.

8. EXPANDING AFTER WELDING

Where expansion after welding is specified, it shall be carried out after the successful completion of the low pressure pneumatic test or leak testing as applicable.

9. HYDRAULIC TESTING

The hydraulic test shall be carried out after all welding, pneumatic tests, leak testing, tube expansion and heat treatment has been completed in accordance with RP 6340 999 0600 005..

TOTAL EXPLORATION PRODUCTION

GENERAL SPECIFICATION

PIPING - VESSELS - WELDING

SP - TCS - 611

WELDING OF PRESSURE

CONTAINING

PIPING AND EQUIPMENT

6	July 92	Modif para. 10 and table 2 and general review
5	November 87	New issue
4	June 87	New issue
Rev. 0	November 80	First issue

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1 - SCOPE

This Specification covers minimum requirements for welding, heat treatment, and non-destructive examination of pressure containing piping and/or equipment such as : pressure vessels, heat exchangers, etc. These requirements also apply to welds attaching skirts, brackets, lugs and other non-pressure parts to such equipment. It is intended to apply to works on fabrication sites such as construction sites or fabrication yards, as well as in manufacturer's workshops.

This document is intended to be used together with ASME and ANSI B31.3 Codes, as applicable.

Welding of the following materials shall comply with both this document and the following related specification :

- Duplex stainless steel (UNS 31803) :
  SP-TCS-614 ("Welding of duplex stainless steels")
- Austenitic Stainless Steel, high Molybdenum (UNS 31254) :
  SP-TCS-615 ("Welding of UNS 31254 alloy")
- Titanium (piping only) :
  SP-TCS-616 ("Welding of titanium pipes and fittings")
- Cupronickel (piping only) :
  SP-TCS-620 ("Specification for fabrication and welding of piping systems in cupronickel and associated materials")
- SP-TCS-617 ("Welding of 3 1/2 % Ni Steel")

2 - DEFINITION OF CONTRACTING PARTIES

OWNER	The legal owner of what is built.
COMPANY	The party which signs the contracts and orders on behalf of the OWNER.
TOTAL	TOTAL acts as the COMPANY representative and performs the role of MANAGER.
ENGINEERING COMPANY	This term designates the engineering companies which work under the supervision of TOTAL and assume the function of Engineer. In all cases a contract exists between the COMPANY and the ENGINEERING COMPANY specifying the missions entrusted to the latter.

MANUFACTURER	The Company responsible for fabrication and installation of piping (either in its facilities or on site), or the Company responsible for the construction of an equipment in its facilities (workshop).
INSPECTION AGENCY	This term designates the inspection bodies which work under the supervision of TOTAL. In some cases, TOTAL may also act as the INSPECTION AGENCY.

3 - CONTRACTUAL REFERENCE DOCUMENTS

3.1 - General

All welding, inspection and heat treatment shall conform as a minimum to the applicable provisions of this Specification, to the requirements of Project Specifications and ENGINEERING COMPANY specifications for specific equipment.

Any construction code or Government regulations which may be listed in the Project Specifications, in the requisition, or in the Purchase Order shall have precedence except where this Specification is more stringent.

Any divergence between any of the Contractual Reference Documents, or between this specification and any other Contractual Reference Document, shall be reported to TOTAL for decision. In such a case, and unless otherwise agreed or decided by TOTAL, it is understood that the more stringent requirements shall apply.

Welds in sour wet condition shall comply with both the present document and the TOTAL specification SP-TCS-612 : "Metallurgical requirements for pressure piping and equipment in sour wet service".

3.2 - Applicable codes and standards

The following publications are referred to herein :

a) ANSI (American National Standards Institute) Standard :

B31.3 - Petroleum Refinery Piping
(Reference herein to ANSI B31.3 applies to piping only)

b) ASME (American Society of Mechanical Engineers) Standard :

Section II	-	Part C : Welding Rods, Electrodes and Filler Metals
Section V	-	Non-destructive Examination
Section VIII	-	Unfired Pressure Vessels, Divisions 1 & 2
Section IX	-	Welding Qualification

c) AWS (American Welding Society)

A5-1 - Specification for Carbon Steel Covered Arc Welding Electrodes
A5-5 - Low Alloy Steel Covered Arc - Welding Electrodes
A5-9 - Specification for corrosion resisting chromium and chromium - nickel steel bare and composite metal cored and stranded arc welding electrodes and welding rods.

The applicable issues are the latest revisions at the date of call for bid.

4 - WELDING QUALIFICATIONS

4.1 - Welding procedures specifications (WPS)

a) The MANUFACTURER shall submit in his tender evidence of his ability to qualify welding procedures for the particular material proposed for each equipment or piping. This evidence shall be based on previous experience and qualification test results.

b) All WPS shall be in writing and shall be qualified and certified in accordance with the applicable code. Such WPS shall not have been in effect for more than six years prior to the date of the purchase order for the equipment being fabricated. All supporting PQR shall be witnessed, or at least certified, by a recognized inspection authority.

c) All WPS, including their qualification test records (P.Q.R.) shall be submitted to the ENGINEERING COMPANY for approval prior to starting fabrication, and this submission shall be considered as part of the calculation and fabrication drawing review by the ENGINEERING COMPANY. All test results shall be from specimens representative of the final condition of the equipment, including heat treatment(s) where applicable.

All materials used for welding procedure qualification tests on carbon steel materials shall be in the upper range of carbon content and carbon equivalent and within the range of alloy elements and mechanical properties of the relevant specifications to be used.

d) Each WPS shall be identified by a unique number. For pressure vessels and other equipment, each WPS shall also be identified on a welding map.

e) The WPS shall state the actual consumable or flux trade name and type as well as the code designation. Procedure tests for all welding processes shall be qualified using the same welding consumables and flux as will be used on the actual item. A change from one Manufacturer's consumable or flux to that from another Manufacturer, or change of type or grade from the same Manufacturer, shall require requalification.

f) Each WPS shall contain complete data including all welding variables with the maximum and minimum values qualified, preheat temperatures, heat treatment details. For submerged arc welding, changes in speed or heat input outside the range qualified shall require requalification.

g) Procedure qualification tests for carbon steel or nickel alloy ferritic steel materials shall include a cross-section micro examination and a hardness survey on a Vickers Pyramid Hardness Testing machine employing a 5 kg load. Surveys shall be made across the complete joint including base metal, heat affected zones and weld metal, 1 mm from the top and 1 mm from the bottom of the joint. Maximum value shall not exceed 350 HV5 unless more stringent requirement due to particular service conditions such as sour wet service is specified otherwise.

h) For inert gas tungsten arc (GTAW) and inert gas metal arc welding (GMAW), the qualification records shall include the composition and flow rate of inert gas shielding and backing when used. For submerged arc welding (SAW), the qualification record shall include the current arc intensity and voltage, and the travel speel for each weld run.

For gas metal arc process (GMAW), the records shall include the power sources, inductance setting, electrode diameter and extension, current and voltage, wire feed rate and travel speed.

i) For low temperature service, -30°C and below, procedure qualifications for ferritic materials and for austenitic stainless steel materials shall include impact tests, whether impact tests are or are not required for the parent metal. Three sets of three specimens each shall be taken at the following locations : weld metal, fusion line and 2 mm from fusion line in the Heat Affected Zone.

For material thicknesses over 38 mm, all the sets of specimens as specified hereabove shall be located on two positions through the thickness :

Position 1 : near (within 1.5 mm) the surface of one side of the material.
Position 2 : midway between the surface and the center of thickness, on the opposite side as described above.

Tests shall be performed in accordance with the relevant Code.

Minimum acceptance criteria shall be in accordance with the applicable Code and/or specification. The lateral expansion shall be reported.

When criteria are not specified, the average Charpy "V" notch impact values in Joules shall be the greather of :

$\frac{Re}{10}$ or : 30 Joules

Where Re = Minimum yield strength of base material (N/mm2) (nominal value)

The minimum value of an individual specimen shall not be less than 80% of the average value specified above.

All impact tests shall be from specimens taken from samples that represent the final condition of the equipment, including heat treatment(s) where applicable. Test temperature for all impact tests shall be at the minimum design temperature unless otherwise specified in the Project Specifications or in the design code.

j) Tack welds shall be made to a qualified welding procedure that has been approved by the ENGINEERING COMPANY. Tack welds shall be completely removed prior to welding. Fit-up lugs and strong backs and other temporary attachments shall be attached using a qualified procedure approved by the ENGINEERING COMPANY.

k) Procedure qualifications for weld overlay deposits shall include the complete chemical analysis of the overlay. Specimens for chemical analysis shall be taken on the final weld layer. The weld metal chemistry shall be within the nominal range specified for the alloy.

For weld overlay deposits made of austenitic stainless steel material of the 300 series of the ASTM, the ferrite content shall be in the range 5% to 15% as determined from the chemistry analysis and with reference to the Schaeffler/Delong diagrams.

The procedure qualification tests shall include :

- dye penetrant examination of the completed weld,
- side bend tests and longitudinal face bend tests for weld metal soudness examination. Excessive fissuring shall be cause for rejection. Fissures shall not exceed four per specimen nor shall they exceed 1.6 mm in length. Cracks corners are not considered,
- photomacrograph.

4.2 - Welders and welding operators

Welders and welding operators shall be qualified in accordance with ASME, Section IX, prior to fabrication. Records shall be submitted to the INSPECTION AGENCY for approval. Alternative qualifications may be accepted subject to the INSPECTION AGENCY approval.

Welding operators and welder qualifications shall normally be valid for one year from qualification performance date.

Welders and welding operators are subject to requalification when the quality of their work, during fabrication, appears to be below the requirements of this Specification.

4.3 - Welding production tests

When required, production tests representative of each material and welding procedure for main seams shall be performed according to contractual documents requirements.

Test coupons shall be of sufficient length to allow for all mechanical tests to be taken.

Test coupons shall be welded attached to the part being welded where possible and shall reflect the actual welding and heat treatment conditions of the piping or equipment.

Mechanical tests and acceptance criteria shall be the same as required for welding procedure qualifications.

5 - WELDING CONSUMABLES

5.1 - Filler materials and fluxes

5.1.1 - Selection of filler materials and fluxes

Welding consumables or combination of them used for the work shall produce welds with strength and ductility properties (especially notch toughness) in the final heat treated condition (if applicable) at least equal to the minimum specified equivalent properties of the base material. For welding carbon steel materials with specified yield strength of 65000 psi and above, the selected consumables shall produce welds with tensile strength property not exceeding by more than 25% the specified minimum tensile strength properties, measured at the time of welding procedure qualification tests on an all-weld tensile test specimen taken from the weld metal.

a) Each heat, lot or batch of consumables shall be subject to acceptance tests and/or inspection and approval by the INSPECTION AGENCY. Mill test certificates showing chemical and mechanical properties shall be established and shall be available on the site at the time of welding. Any consumables deemed unsuitable for use shall be removed from the site.

b) Filler metal for welding similar materials shall be as shown in Table 1. 9% Ni materials shall be welded using filler materials complying with one of the following classifications of the AWS A 5.11 : ENi Cr Fe - 3, or ENi Cr Mo - 3. Guaranteed strength levels are required for mechanical design.

c) Filler metals for welds joining dissimilar materials shall be in accordance with Table 2. Filler metals for combination of materials other than those shown in Table 2 shall be submitted to TOTAL for approval.

   Welds joining carbon steel materials of different grades shall give the same strength as that specified for the higher grade of material and shall have ductility and notch toughness properties equal to the higher values specified for the grades of steel being joined.

d) In welding processes other than SMAW or FCAW, the bare filler wire shall contain all the alloying elements ; no elements shall be added via the flux. Consumables for carbon steel materials shall not contain more than 2.5% of alloying elements.

e) All welding products shall be used within the limits recommended by their Manufacturer and the welding variables used for fabrication shall be within the range used for the procedure qualification.

5.1.2 - Supply

Low hydrogen electrodes and fluxes shall be supplied in sealed moisture proof containers.

Each heat, lot or batch of consumables shall be subject to acceptance tests and/or inspection and approval by the INSPECTION AGENCY.

Mill test certificates showing chemical and mechanical properties shall be established and shall be available on the site at the time of welding. Any consumables deemed unsuitable for use shall be removed from the site.

5.1.3 - Storage

Electrodes, filler wires and fluxes shall be kept clean, dry and properly stored according to Manufacturer's recommendations. No electrodes, filler wires or fluxes that are damaged, damp, greasy or oxidized may be used.

MANUFACTURER's facilities shall include a temperature and humidity controlled welding consumable and flux store, including holding and drying ovens. Storage of electrodes, wire and flux in sealed containers shall be in accordance with Supplier's instructions. Storage of open containers shall be in accordance with Supplier's instructions but at a minimum temperature of 75°C in any case.

Low hydrogen electrodes and fluxes shall be placed in a holding oven and held at a minimum temperature of 150°C for at least one hour prior to use. Welding electrodes shall be issued for production from holding ovens only and shall be placed in heated quivers capable of maintaining a minimum temperature of 70°C. After four hours, the remaining unused electrodes in the quivers shall be placed in a drying oven and held at 250 - 300°C for hour hours or as recommended by the Supplier if at a higher temperature, and then transferred to a holding oven prior to reissue.

Low hydrogen electrodes which have been in direct contact with water shall be definitely rejected and removed from the site.

5.2 - Backing rings and consumables inserts

Permanent backing rings and consumable inserts shall not be used. Temporary backing rings may be used after approval of the ENGINEERING COMPANY and only if they are made of the same material that has to be welded ; in such a case, at least the two first layers of passes shall be gouged out after completion of the weld, and rewelded. Other technique shall require specific TOTAL prior approval.

TABLE 1

Filler metal for welds joining similar materials

Base Material		Covered Electrodes			Welding Rods	
		ASME Spec	Preferred Electrode	Alternative Electrode	ASME Spec	Classification
Carbon Steel	(1)	SFA5.1	E70XX (3, 4, 9)	E60XX (3, 4)	(5, 6, 7, 8)	-
	(2)	SFA5.1	E70XX (3, 4, 9)	-	(5, 6, 7, 8)	-
Carbon-molybdenum		SFA5.3	E70XX-A1	-	(5)	-
ASTM A204 Gr. A or B		SFA5.5	E70XX-A1	-	(5, 11)	-
ASTM A302 Gr. A or B		SFA5.5	E9018-D1 (0.41 Mo min)	E9015-D1 (0.41 Mo min)	(5, 11)	-
2 1/4 nickel		SFA5.5	E8018-C1	E8016-C1	(5, 11)	-
3 1/2 nickel		SFA5.5	E8018-C2	E8016-C2	(5, 10, 11)	-
Low Chromium Ferritic Steels	1/2 Cr - 1/2 Mo	SFA5.5	E8018-B2L	E8015-B2L (17)	(5, 11, 15, 17)	0.05 C max
	1 Cr - 1/2 Mo	SFA5.5	E8018-B2L	E8015-B2L (17)	(5, 11, 15, 17)	0.05 C max
	1 1/4 Cr - 1/2 Mo	SFA5.5	E8018-B2L	E8015-B2L (17)	(5, 11, 15, 17)	0.05 C max
	2 1/4 Cr - 1 Mo	SFA5.5	E9018-B3L	E9015-B3L (17)	(5, 11, 15, 17)	0.05 C max
	5 Cr - 1/2 Mo	SFA5.4	E502-15 0.05 C max	(17)	SFA5.9 (17)	ER 502 0.05 C max
	7 Cr - 1/2 Mo	SFA5.4	E7Cr-15 0.05 C max	(17)	(5, 17)	0.05 C max
	9 Cr - 1 Mo	SFA5.4	E505-15 0.05 C max	(17)	SFA5.9 (17)	ER505 0.05 C max
Ferritic Stainless Steels	AISI Type 405	SFA5.4	Inco-Weld A Inconel 182	E410-15 0.05 C max (16)	Inconel 82 or SFA5.9	ER410 0.05 C max
	AISI Type 410S	SFA5.4	Inco-Weld A Inconel 182	E410-15 0.05 C max (16)	Inconel 82 or SFA5.9	ER410 0.05 C max
	AISI Type 410	SFA5.4	Inco-Weld A Inconel 182	E410-15 0.05 C max	Inconel 82 or SFA5.9	ER410 0.05 C max
	AISI Type 430	SFA5.4	E430-5	-	SFA5.9	ER430
Austenitic Stainless Steels	AISI Type 304	SFA5.4	E308-15	E308-16	SFA5.9	ER308
	AISI Type 304L	SFA5.4	E308L-15	E308L-16	SFA5.9	ER308L
	AISI Type 321	SFA5.4	E347-15	E347-16	SFA5.9	ER347
	AISI Type 347	SFA5.4	E347-15	E347-16	SFA5.9	ER347
	AISI Type 316	SFA5.4	E316-15	E16-8-2-15	SFA5.9	ER316
	AISI Type 316L	SFA5.4	E316L-15	E316L-16	SFA5.9	ER316L
	AISI Type 309	SFA5.4	E309-15	E309-16	SFA5.9	ER309
	AISI Type 310	SFA5.4	E310-15	E310-16	SFA5.9	ER310
	Alloy 800, 800H	SFA5.11	(12)	-	SFA5.14	(12)
Nonferrous metals and alloys	Aluminium and Aluminium Alloys	SFA5.10	-	-	SFA5.10	-
	Aluminium bronze	SFA5.6	EcuAl-A2 (18)	-	SFA5.7 (18)	ERCuAl-A2 (18)
	Phosphor bronze	SFA5.6	ECuSn-A	ECuSn-C	SFA5.7	ERCuSn-A
	Copper	SFA5.6	ECu	-	SFA5.7	ERCu
	67 Ni-30Cu	SFA5.11	ENiCu7	-	SFA5.14	ERNiCu-7
	Hastelloy	-	(13)	-	-	(13)
	Alloy 600	-	(12)	-	-	(12°
	70Cu-30Ni - 90Cu-10Ni	SFA5.6	ECuNi (18)	-	SFA5.7 (18)	ERCuNi (18)
	Nickel	SFA5.11	ENi-1	-	SFA5.14	ERNi-1
Miscellaneous	20 Cr - 29 Ni 2 1/2 Mo - 3 Cu	SFA5.4	E320-15 (14)	-	SFA5.9	ER320 (14)

(Numbers in parentheses refer to notes hereafter)

Notes to Table 1
(Numbers in parentheses in Table 1 refer to the following notes :)

(1) Minimum ultimate strength less than 450 N/mm2.

(2) Minimum ultimate tensile strength 450 N/mm2 and greater.

(3) Low hydrogen electrodes and fluxes, or a low hydrogen weld process shall be used where any of the following requirements or conditions apply :

a) impact testing of weld metal
b) the carton content of the base material exceeds 0.22%
c) the base material thickness exceeds 12 mm
d) the specified yield strength of the base material exceeds 260 N/mm2

(4) The following classifications are not acceptable for use in pressure containing welds : E6012, E6013, E6022, EXX14 and EXX24. However, these classifications may be used for tank roof and bottom fillet welds after prior approval of TOTAL. For single sided joints, E6010 and E6011 electrodes are permissible for root passes on piping only.

(5) Where no ASME material specification exists for wire or rods, wire or rods of the same nominal composition as the base material with substantially neutral flux or inert gas may be used if they have been qualified in accordance with the applicable code and specification requirements.

(6) Gas metal arc welding (GMAW) wire shall conform to ASME-5.18 and SFA-5.20.

(7) Submerged arc welding (SAW) wire and flux shall conform to ASME SFA-5.17 or SFA 5.23 ; however, equivalence under these standards shall not be considered adequate for substitution between different manufacturers or between a single manufacturer's grades without requalification.

(8) For gas tungsten arc welding (GTAW), ASME SFA-5.18 ER70S-2 is the perferred welding wire.

(9) ASME SFA-5.1 E6010 welding electrode is acceptable for the root pass in piping welding only, if the procedure has been qualified to the appropriate code and specification requirements.

(10) Alternatively, E7016-C2 welding electrodes or austenitic filler materials may be used dependant upon the minimum design temperature and toughness requirements.

(11) Submerged arc welding (SAW) wire and flux shall conform to ASME SFA-5.23. Wire and flux combinations shall deposit welds with equivalent composition and equivalent mechanical properties as the base material. Limitations on substitution shall be as specified in Note 7 above.

(12) For design temperatures less than 535°C, ENiCrFe-3 or ERNiCr-3 shall be used. For design temperatures of 535°C to 815°C, ENiCrFe-2 or ERNiCr-3 shall be used. For design temperatures greater than 815°C, ENiCrMo-3 or ERNiCrMo-3 shall be used.

(13) Welding rods or electrodes used shall require TOTAL advance approval, and shall deposit weld material of mechanical and corrosion resistant properties equivalent to those of the base material.

(14) Type 320 (low residual) filler metal is an acceptable alternative when weld hot-cracking is considered a potential problem.

(15) For gas tungsten arc (GTAW) and gas metal arc (GMAW) welding processes, rods and wire shall conform to ASME SFA-5.28).

(16) Type E309-15, E309-16 or ER309 with 0.065 minimum carbon content may be used in some design conditions with advance approval of TOTAL (not to be used in cyclic service).

(17) For chromium-molybdenum steel, other filler materials such as Inconel 82 and 182, or Inco-Weld A, may be used for specific services, subject to TOTAL approval.

(18) For information only, as for cupronickel and associated materials, filler material shall comply with TOTAL specification SP-TCS-620 "Specification for fabrication and welding of piping systems in cupronickel and associated materials". Brazing (silver brazing) may also be considered, and, according to the specification SP-TCS-620, is mandatory in some circumstances.

TABLE 2

Filler metal for welds joining dissimilar materials

Base Material Number	Base Material type	Base Material Number															
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	Carbon Steel	A	D	B	B	B	B	B	B	B	B	B	B	B	B	B	B
2	Carbon-Molybdenum Steel		D	B	B	B	B	B	B	B	B	B	B	B	B	B	B
3	3 1/2 % Nickel Steel			E	B	B	B	B	B	B	B	B	B	B	B	B	B
4	9 % Nickel Steel				B	B	B	B	B	B	B	B	B	B	B	B	B
5	AISI Type 410 S					B	B	B	B	B	B	B	B	B	B	B	B
6	AISI Type 410						B	B	B	B	B	B	B	B	B	B	B
7	AISI Type 304							G	G	H	H	H	H	H	H	B	B
8	AISI Type 304 L								K	H	H	H	H	H	H	B	B
9	AISI Type 321									H	H	H	H	H	H	B	B
10	AISI Type 347										H	H	H	H	H	B	B
11	AISI Type 316											L	L	C	N	B	B
12	AISI Type 316 L												M	C	N	B	B
13	AISI Type 309													C	N	B	B
14	AISI Type 310														N	B	B
15	Incoloy 825															B	B
16	Inconel 625																B

Filler material AWS classification :

A - E-XX16 or E-XX18

B - Inco-Weld A, Inconel 82 or Inconel 182

C - E309-15 or E309-16

D - E7018-A1

E - E80XX-C1. Alternatively one of following classifications of the AWS A.5.11 : E Ni Cr Fe-3 or E Ni Cr Mo-3, subject to approval by TOTAL, may also be used

G - E308-15 or E308-16

H - E347-15 or E347-16

J - Use filler materials complying with one of the following classifications of the AWS A.5.11 : E Ni Cr Fe-3 or E Ni Cr Mo-3

K - E308L-15 or E308L-16

L - E316-15 or E316-16

M - E316L-15 or E316L-16

N - E310-15 or E310-16

6 - JOINT PREPARATION, SPACING AND ALIGNMENT

6.1 - Edge preparation

a) Weld bevels shall be suitable for the welding process to be used and shall be as defined in the applicable qualified welding procedure. The bevel root gap and offset of butted edges shall be as required by the applicable construction code. Whenever necessary, spacing tools shall be used to ensure proper root opening. Double sided weld joints must have the root pass entirely removed by arc-air gouging, followed by grinding for final cleaning, prior to back-welding. Double sided weld joints shall be used whenever possible ; joints designed in single-welded butt joints shall be such that full penetration is attained.

b) Removable starting and stopping tabs shall be used for longitudinal welding where automatic welding processes are used.

c) Weld bevels shall be made by machining, grinding or thermal cutting, and the surfaces shall be reasonably smooth and true. For Aluminium and Aluminium alloys, the final surfaces for welding on shall be finished by machining or filing (not grinding).

d) Materials which require preheat for welding shall be preheated in the same manner for thermal cutting or arc-air gouging.

e) When arc-air gouging is used, the surface shall have all carburized and hardened surfaces removed by grinding, and all such areas shall be inspected by the MANUFACTURER's quality control department for conformity prior to welding.

f) All surfaces and edges to be welded shall be smooth, uniform and free from cracks, tears, gouges and other discontinuities which could adversely affect the quality or strength of the weld.

6.2 - Cleaning

a) Surfaces to be welded shall be clean and free from paint, oil, dirt, scale, oxides and other materials detrimental to welding. Cleaning shall be done in a manner that will not lead to contamination of the weld or base metal. Only stainless steel brushes and tools shall be used on stainless steels, nickel and non ferrous materials. Grinding discs containing sulphur (iron sulphite) or other harmful components shall not be used on stainless steels, nickel or non ferrous materials.

Upon completion of each welding pass, the weld shall be cleaned of spatter, slag and flux deposits.

b) For Aluminium or Aluminium alloys, each layer of welding shall be cleaned free of oxides and other foreign material, utilizing stainless steel wire brushing, before depositing the next weld beads. Each layer of weld shall be free from irregularities such as high spots, deep crevices, undercutting and porosity. The above can be achieved by filing or chipping and stainless steel wire brushing.

Immediately prior to setting-up and welding, the bevel preparation and at least 12 mm back from the bevel, both internally and externally, shall be stripped free of oxide film by means of filing or stainless steel wire brushing, and further cleaned free of dirt, grease, etc..., using a clean cloth or paper towel and a non-toxic solvent. Disc grinding is not permitted. All tools used, i.e. scrapers, wire brushes, files etc, must be used only on aluminium to prevent contamination by other metals and materials.

Filler rods for inert gas tungsten arc welding shall be cleaned immediately before use. Electrode wire for inert gas metal arc welding shall be protected from contamination during use and in particular between non-working periods.

6.3 - Tack welds

All tack welds shall be made in accordance with a previously approved welding procedure and shall be performed by qualified welders. Tack welds shall be of sufficient cross-section and length in order to avoid cracks, especially on high strength steel materials.

6.4 - Temporary Welds

Temporary welds for temporary handling attachments, lugs, etc... shall be minimized. When temporary welding is deemed necessary, welding shall be performed with the same care, materials, electrodes, minimum preheat and procedures as for the main fabrication welding.

The temporary welds shall subsequently be removed and ground flush with the base material, and then inspected by non-destructive methods (Magnetic Particle or Dye Penetrant). On non-magnetic materials, this examination shall be made by the dye-penetrant method.

Any defect found during this inspection shall be removed and repaired by a qualified welder using welding procedures approved by the ENGINEERING COMPANY. The method of removing attachments shall not injure the metal surface (i.e. they shall not be removed by hammering off).

7 - PRODUCTION WELDING

7.1 - General requirements

a) Production welding may only start when qualified welding procedures and welders have been approved by the ENGINEERING COMPANY and the INSPECTION AGENCY.

b) Preparation for welding, types and sizes of electrodes, current amperages and voltages used, preheating requirements etc... shall be the same for production welding as those defined in the applicable qualified welding procedures.

c) Adequate precautions shall be taken and suitable equipment shall be available on site in order to protect the welds (and welders) from adverse weather conditions (rain, wind,...) at the time of welding. If weather conditions become too severe, welding operations shall be stopped. No welding shall be done on wet base materials.

d) All surfaces to be welded shall be visually inspected : they shall contain no laminations or other injurious defects.

e) Peening of welds is not permitted.

f) Structural attachment welds shall be continuous (skip welding not permitted). Fillet welds shall have a minimum leg length of 4 mm.

g) Cleanliness shall be maintained after completion of welding. All stubs, rods, flux and foreign materials shall be removed from the equipment or piping.

h) Stainless steel surfaces shall be restored to their corrosive resistant state. Chemical cleaning procedures, if applicable, shall be subject to TOTAL approval.

7.2 - Welding processes

7.2.1 - Approved welding processes

All welds shall be made by either the shielded metal arc (SMAW), inert gas tungsten arc (GTAW), inert gas metal arc (GMAW) or submerged arc (SAW) processes, except that for Aluminium and Aluminium alloys, only the argon GTAW or GMAW processes shall be used. Any other welding process such as FCAW (Flux Cored Arc Welding process) requires TOTAL prior approval. The oxy-acetylene process may be used for carbon steel piping (excluding valves) in nominal diameters 1 1/2" and smaller with wall thickness of 7.6 mm and less and on utility piping systems only.

Single-side welded butt joints in carbon-molybdenum, chromium-molybdenum, and austenitic stainless steels, and welds located in compressor suction or lub oil systems, shall have the root pass made by the GTAW or GMAW process with an appropriate back gas shielding protection. When the GTAW or GMAW processes are used for welding carbon, carbon-molybdenum and low-alloy chromium molybdenum steels with a chromium content not exceeding 1.5%, in other applications than above, the requirement for back gas shielding may be waived, subject to TOTAL prior approval.

7.2.2 - Limitations and requirements

a) SMAW

Use of SMAW in the downhill direction is restricted to utility lines ; any other vertical welding shall be done vertical up.

The maximum width of weave shall not exceed two times the electrode diameter.

b) SAW

The maximum individual layer thickness for submerged arc welds shall not exceed 9.5 mm for materials less than 32 mm thick, or 12.5 mm for thicker materials. Recycling of fluxes shall not be permitted.

The brand and grade of flux used for the production of submerged arc welds shall be same as was used in the procedure qualification test. Equivalent fluxes made by other Manufacturers will only be permitted provided they have been also qualified in an appropriate P.Q.R. All alloy additions must be made through the filler metal.

c) GMAW

Inert gas metal arc welding with solid wire electrodes shall only be used in the spray transfer current range.

d) GTAW

Thoriated tungsten shall not be used for the non-consumable electrode when welding Aluminium or Aluminium alloys.

7.3 - Welding technique and sequences

7.3.1 - Welding technique

Welding techniques shall be selected to assure that specified tolerances for straightness and out-of-roundness are not exceeded. If such tolerances are not stated in the drawings, standards or specifications, the applicable section of the relevant code shall govern.

7.3.2 - Welding sequences

The MANUFACTURER shall provide and exert all necessary supervision to ensure that the planned sequences are observed.

a) The sequences in assembling, joining and welding the various parts to be welded shall be scheduled in order to minimize distortion, warpage, shrinkage and accumulations of residual joint stresses in each part of the piping systems or of the equipment being welded.

b) Insofar as practicable, all welds shall be deposited in a sequence that will balance the applied heat of welding while welding is in progress.

c) The direction of the general progression in welding parts of piping systems shall be from points where the parts are relatively fixed in position with respect to each other towards points where they have a greater relative freedom of movement.

d) Joints expected to have significant shrinkage should usually by welded before joints expected to have lesser shrinkage. They should also be welded with as little restraint as possible.

In making welds under conditions of severe external shrinkage restraint, the welding shall be carried-out continuously to completion or to a point that will insure freedom from cracking before the joint is allowed to cool below the minimum specified preheat and interpass temperatures.

e) Where residual joint stresses are not desirable and cannot be avoided, suitable postweld heat treatments shall be performed. The heat treatments shall be made taking into account the recommendations of this Specification.

f) Welded joints shall be made by completing each layer before the following layers are deposited.

7.3.3 - Heat input control

When welding quenched & tempered steels, or ferritic steels impact tested at -46°C or below, the heat input, in conjunction with the maximum preheat and interpass temperatures, shall be restricted to the maximum values shown in the relevant WPS and PQR. The above limitations shall be in strict accordance with the steel producer's recommendations.

The following parameters shall be carefully checked :

- Preheat temperature
- Minimum and maximum interpass temperatures
- Amperage and voltage
- Welding travel speed

The MANUFACTURER shall have available at the work site suitable equipment to measure these important variables (eg. contact thermometer, etc.).

Accuracy of parameters readings :

- Temperature $\pm$ 10°C
- Amperage and voltage $\pm$ 5%

7.4 - Weld contour and finish

7.4.1 - Flux and slag removal

Weld beads shall be contoured to permit complete fusion at the sides of the bevel and to minimize slag inclusions. Flux, slag, and weld spatters shall be completely removed from weld beads and from the surface of completed welds and adjoining base material. The flux removal shall be done in a manner that will not contaminate or overheat the weld or adjoining base material.

7.4.2 - Weld reinforcement

Weld reinforcement and finish shall be as required by the applicable ASME or ANSI standard, except that undercutting of the base metal shall not be permitted. Internal protrusion shall not exceed the reinforcement thickness specified by the applicable standard. Weld finish shall not impair the interpretation of NDT results. Code requirements regarding absence of coarse ripples, grooves, overlaps, abrupt ridges and valleys shall be strictly conformed to.

The geometrical unsharpness shall not exceed 0.2 mm unless otherwise approved by the INSPECTION AGENCY in particular cases.

7.4.3 - Weld surfaces

Weld surfaces shall be free of cracks, excessive porosity, slag inclusions and other defects indicative of poor workmanship.

7.4.4 - Arc strikes

Arc strikes outside weld bevels shall be avoided. Should this occur, however, the deposit shall be carefully removed by mechanical means and the area shall be 100% examined by magnetic particle (on magnetic materials) or dye penetrant (on non magnetic materials). Repairs shall be in accordance with this Specification.

7.5 - Corrosion resistant overlays

7.5.1 - For weld overlays, there shall be no less than two layers of corrosion-resistant weld metal.

For stabilized or low-carbon 18/8 type Cr-Ni austenitic stainless steel cladding, the first layer of deposited weld metal shall be 25/12 type Cr-Ni austenitic stainless steel which shall be niobium stabilized or low-carbon.

Subsequent layers of deposit shall be niobium stabilized 18/8 type Cr-Ni austenitic stainless steel unless otherwise specified or agreed.

7.5.2 - When joining integrally clad plate :

- the clad layer shall be stripped for a minimum distance of 10 mm from the bevel and the base material shall be etched with nitric acid to assure complete removal of cladding material,
- removal of cladding shall not reduce the base material thickness below the design thickness,

- preparations of local repair cavities in overlay welds which penetrate the base material more than 10% or 3 mm, whichever is the smaller, shall have the base metal rewelded with a procedure and materials consistent with the welding of the base materials.

7.5.3 - For plate with applied alloy lining, base plate welds that would be exposed to the corrosive fluid of the equipment shall be overlayed according to paragraph 7.5.1. The weld detail shall be as shown on approved drawings.

8 - PREHEATING

The following requirements shall apply in addition to those specified in the applicable ASME or ANSI B31.3 standard. In case of discrepancy, the more stringent requirement shall apply, unless otherwise agreed by TOTAL prior to qualification of the welding procedures. Preheat temperatures higher than those specified herein may be required during the qualification tests in order to meet contractual requirements such as hardness levels, or to prevent from cold cracking (particularly in the case of highly restained welds).

The preheating which has been qualified in the applicable WPS shall apply not only to welding, but also to all thermal operations such as thermal cutting, arc-air gouging, and also tack-welding.

In weld assembly of dissimilar or different base materials, preheat temperature shall be the highest of those which are requested for each material.

a) For carbon steel and low alloy carbon steels, preheat temperatures shall not be less than :

50°C When only moisture removal is required. When moisture removal is not applicable, minimum temperature of metal shall be 0°C for low hydrogen electrodes and 10°C for other types.

100°C When the base material thickness exceeds 25 mm, or when the carbon equivalent exceeds 0.420.

150°C If both above conditions are met, or when any integrally reinforced nozzle greater than 25 mm thick is being welded into a vessel shell or head.

Note : Carbon equivalent CE is to be calculated from the IIW formula :

$$CE = C + \frac{Mn}{6} + \frac{Cr + Mo + V}{5} + \frac{Ni + Cu}{15}$$

b) Preheat temperatures for other ferritic materials shall be as shown in Table 3.

TABLE 3

Minimum preheat temperatures for ferritic materials

Material (Nominal analysis)	ASME P-N°	Minimum preheat temperature (Degrees Celsius)
Manganese - Molybdenum	3	150
C - 1/2 Mo	3	100
1/2 Cr - 1/2 Mo	3	100
1 Cr - 1/2 Mo	4	150
1 1/4 Cr - 1/2 Mo	4	150
2 1/4 Cr - 1 Mo	5	200
3 Cr - 1 Mo	5	200
5 Cr - 1/2 Mo	5	200
7 Cr - 1/2 Mo	5	200
9 Cr - 1 Mo	5	200
2 1/2 Ni	9A	150
3 1/2 Ni	9B	150 (1)
5 Ni	11A	150
9 Ni	11A	150

Note (1) : for piping welding, preheat at 95°C only is acceptable for thicknesses up to 10 mm.

c) For Aluminium and Aluminium alloys, preheat shall be as follows :

- Prior to welding, the joint shall be heated uniformly to a temperature of 90°C but not more than 120°C.

- The above preheat temperature shall be maintained during all welding operations. On heavy sections, it is important that preheat be allowed to "soak" evenly throughout the base metal, so that the through metal temperature is 95°C - 120°C. It is equally important that these temperatures (for preheat and interpass) are not exceeded.

- Should the welding be interrupted, the joint preheat temperature shall be maintained if possible. In any case, cooling under insulation is required, and re-heating at original temperature is necessary before welding resumes.

d) For ferritic materials, when the required preheat temperature is 200°C or higher, and for thicknesses greater than 50 mm, the metal shall be maintained at preheat temperature until the weld is postweld heat treated. Alternatively, the weld shall be given a hydrogen diffusion treatment before allowing any cooling. This treatment shall consist of heating to 250°C minimum, holding for four hours, then slowly cooling under thick dry insulation blankets to ambient temperature. For lower preheat temperatures, the joint shall have been completed and the cooling rate minimized by wrapping the joint in dry insulating blankets.

Should the welding be interrupted, the required minimum preheat temperature shall be re-established over the entire joint length to be welded before welding resumes.

e) For austenitic stainless materials, preheating is not required.

f) Interpass temperatures shall not be greater than the preheat temperatures or the highest interpass temperature recorded during the welding procedure qualification test.

g) The required minimum preheat temperature shall be applied to the whole length of joint to be welded and shall be maintained until the weld is completed. Application of preheat when using gas burners shall avoid deposition of carbon. The minimum preheat temperature shall be established for a sufficient distance either side of the joint to avoid unsatisfactory thermal gradient. This distance shall not be less 75 mm each side of the joint.

h) The preheat temperature shall be measured on the face opposite to that being heated when possible. When this is not possible, allowance shall be made for temperature equalization, e.g. remove heat source and allow period of one minute for each 25 mm thickness of material to elapse, before measuring temperature.

i) Measurement of preheat and interpass temperatures shall be done by temperature indicating crayons, contact pyrometer or other approved method of temperature measurement. Whatever method to be used shall be stated on the welding procedure qualification and subject to approval by the ENGINEERING COMPANY. Maximum and minimum temperatures shall be specified. Temperature indicating crayons shall not be applied to surfaces where weld metal is to be deposited.

9 - POSTWELD HEAT TREATMENT

a) Postweld heat treatment shall be in accordance with the applicable ASME or ANSI B31.3 Code. Additionally, the following shall apply :

- Post Weld Heat Treatment of ferritic materials shall conform to Table 4 as a minimum.
- No Post Weld Heat Treatment shall be performed on Austenitic Stainless Steel materials.
- The postweld heat treatment for welds joining austenitic stainless steels to dissimilar material shall be as specified in the qualified welding procedure and approved by the ENGINEERING COMPANY prior to fabrication.
- Postweld heat treatment of field welds in low chromium-molybdenum steel (P-4 and P-5 groups) made with Inconel 82 or 182 filler metal may be omitted subject to TOTAL approval.

b) Postweld heat treatment procedures and temperatures shall be carefully controlled and recorded. Heat treatment charts shall be submitted to the INSPECTION AGENCY. In case the heat treatment is sub-contracted in a workshop not belonging to the actual MANUFACTURER of the vessel, the MANUFACTURER shall witness the heat treatment operation at least during the period of time when the temperature is above 300°C, and shall sign the heat treatment charts.

TABLE 4

Postweld heat treatment for carbon steels and ferritic alloy materials*

Material (Nominal analysis)	ASME P-N°	Minimum preheat temperature (Degrees Celsius)
Carbon steel	1	600
Manganese - Molybdenum	3	620
C - 1/2 Mo	3	620
1/2 Cr - 1/2 Mo	3	620
1 Cr - 1/2 Mo	4	650
1 1/4 Cr - 1/2 Mo	4	650
2 1/4 Cr - 1 Mo	5	720
3 Cr - 1 Mo	5	720
5 Cr - 1/2 Mo	5	720
7 Cr - 1/2 Mo	5	720
9 Cr - 1 Mo	5	720
2 1/2 Ni	9A	565
3 1/2 Ni	9B	565
5 Ni	11A	565
9 Ni	11A	565

* Does not apply to normalized and tempered or quenched and tempered materials. Postweld heat treatment temperatures of such materials shall not exceed the tempering temperature of the base material and shall be such that the mechanical properties are not less than the minimum specified.

Notes :

1. Temperature of any part of weldment during postweld heat treatment shall not be less than values shown herein. Sufficient thermocouples shall be properly attached to the workpiece so as to accurately indicate metal temperatures reached. Results shall be recorded on a chart.

2. Maximum temperature of any part of weldment during postweld heat treatment shall not be more than 25°C over the minimum values shown herein unless otherwise approved by TOTAL.

10 - INSPECTION AND EXAMINATION

a) Examination of welds shall be in accordance with the procedures and acceptance standards of the applicable ASME or ANSI B31.3 standards, the Project specifications, this Specification, and the references given hereafter in Table 5.

TABLE 5

Weld examination procedures and acceptance standards

Examination method	Applicable standard(s)
Visual	ASME Sect. V, Article 9
Radiography	ASME Sect. VIII, Div. 1 Par. UW 51 ASME Sect. VIII, Div. 2, Art. 1-5 ANSI B31.3, Para. 336.45 (Piping only)
Magnetic particle	ASME Sect. VIII, Div. 1 App. 6 ASME Sect. VIII, Div. 2 App. 9-1
Ultrasonic	ASME Sect. VIII, Div. 1, App. 12 ASME Sect. VIII, Div. 2, App. 9-3
Liquid penetrant	ASME Sect. VIII, Div. 1, App. 8 ASME Sect. VIII, Div. 2, App. 9-2

b) Radiographic inspection :

- Radiographic inspection of welds may be performed with gamma rays for piping only. In all other applications, X-rays shall be used exclusively, unless otherwise agreed in writing by TOTAL.

- Films shall be carefully identified. The markings shall not be placed on welds and shall include weld number, line identification and welder's symbol.

- Fine grain films (or ultra fine grain films) shall be used. The INSPECTION AGENCY may require the use of ultra fine grain films, particularly with gamma rays, where a better quality of the radiographic image is needed.

  For gamma rays inspection, only D4 (AGFA GEVAERT) or KODAK "M" films shall be used. Equivalent or better quality films (i.e. with finer grains) may be proposed as an alternative, subject to TOTAL prior approval.

- Sensitivity of films shall be verified by the use of wire-type Image Quality Indicator (IQI) of DIN standard placed transversely to the weld on the source side, for each exposure. The IQI may be placed on the film side if the radiographic procedure has shown that acceptable sensitivity level may be obtained.

  The sensitivity maximum acceptable values, as a function of thicknesses examined, are given in the following table :

TABLE 6

Image quality indicator sensitivities

Specimen thickness (mm)	IQI sensitivity (%, maximum value)
3	3.0
6	2.5
12.5	2.0
25	1.7
40	1.5
50	1.3
75	1.1
100	1.0
150	0.9

- Single-film viewing shall be used, and the film density shall be in the range of 2.5 to 3.5. For those instances where the variable thickness makes single film impracticable, with the INSPECTION AGENCY approval, double film viewing may be used. For the double film technique, the film density shall be in the range of 2 to 3.5 for the double film combination or each individual film. Higher densities (up to 4.0) may be acceptable if adequate viewing and satisfactory interpretation of higher density film are permitted by the viewing equipment.

- Radiography of welds in pipe having a nominal diameter of 3" or less may be performed by the elliptical projection technique. At least two separate exposures are required at locations 90° apart.

- Where there is no internal access, radiographs of welds in pipe shall be double wall technique with a minimum of 2 shots up to 4" in diameter and 3 shots over 4" in diameter and only that portion of the weld on the film side of the pipe (opposite to the radiation-source side) shall be interpreted. Because of the variation in pipe diameters, wall thicknesses and source-to-film distances, it may be necessary to take more than the minimum number of radiographs to properly examine the entire circumference of a weld.

- Welds of a size or type preventing conclusive radiograph images, (this does not apply to welds in pressure retaining service including nozzle welds), such as some types of branch connections and fillet welds, shall be given a magnetic particle examination. For non-magnetic materials and nickel alloys, a liquid penetrant examination shall be used. Internal surfaces shall be examined where accessible.

c) Magnetic particle examination shall be performed by the electro-magnet method. The prods method shall not be used because this equipment could produce arc strikes on the pieces to be tested.

d) The extent of inspection shall be in accordance with the contractual requirements and shall not be less than the following :

- All welds shall be 100% visually inspected after completion.
- When welds in pressure-containing equipment of chromium-molybdenum materials are subject to 100% radiography, the welds shall also be examined inside (when accessible) and outside by the magnetic particle method after postweld heat treatment.
- The attachment welds between structural components and pressure parts of ferritic alloy materials (P-3 and higher), or on equipment designed for temperatures below 0°C, shall be magnetic particle examined after postweld heat treatment.
- All pressure-containing equipment designed with 100% joint efficiency, irrespective of material, shall have all nozzle and reinforcing pad attachments and attachments to pressure parts welds examined by the magnetic particle or liquid penetrant method. Inspection shall be performed before and after postweld heat treatment both outside and inside where accessible.
- For spot radiographed equipment, this shall not be less than 10% of the weld length.

e) When radiographic examination is required and is not practicable, ultra-sonic examination plus magnetic particle inspection may be considered as an alternate subject to TOTAL approval. Dye penetrant inspection may also be used to replace MPI when deemed necessary, subject to TOTAL approval.

f) For clad constructions, where the thickness of the cladding has not been included in the strength calculation, allowable defect sizes in the cladding may be modified provided the corrosion resistance is not reduced. Such changes must be agreed by TOTAL at the design stage. Acceptability of cladding defects shall be related to the cladding thickness and not to the total wall thickness.

g) Tungsten inclusions in welds (when using the GTAW process) are not permitted and shall be repaired even if small in size or in quantity.

11 - REJECTION AND REPAIR

Defects that are outside the limits of the applicable codes and standards, this Specification, Project Specifications or other requirements stated on the Purchase Order, shall be cause for rejection and the MANUFACTURER shall take such remedial action as is necessary to secure acceptance. Such work shall be subject to the INSPECTION AGENCY approval and shall be at the sole expense of the MANUFACTURER.

Complete repair of a weld shall include removal of the weld, bevelling of new edges, and re-welding.

Weld overlays which are found by inspection to be unsound, or which are deposited by procedures differing from those properly qualified, shall be rejected, completely removed from the equipment, and replaced using an approved procedure.

When welding is judged to be unsatisfactory by the INSPECTION AGENCY, the individual responsible for the work shall be suspended from welding and all his work examined by non-destructive means. Welding found to be unacceptable by the INSPECTION AGENCY shall be repaired. The individual may be re-assigned only after additional training, the completion of satisfactory re-qualification tests, and with the approval of the INSPECTION AGENCY.

All repairs shall be documented and shall be included in the MANUFACTURER's data book. Repairs shall be advised to the INSPECTION AGENCY ; the repair method, the welding procedure and welder qualifications, etc..., shall be in accordance with this Specification and shall be approved by the INSPECTION AGENCY before any repair being initiated. This approval must be based on qualification testing (PQR) supporting a specific WPS for each type of repair on materials other than austenitic stainless steels which are subject to impact tests at temperatures below 0°C.

Welds containing cracks shall be subject to additional non-destructive testing (Ultrasonic or Magnetic Particle) and then, the extent of repair shall be decided by the INSPECTION AGENCY. Repair welding will only be permitted after consideration of the nature and cause of cracking.

Unacceptable defects shall be removed by chipping, grinding, machining or air-arc gouging. Where air-arc gouging is used, all carbon, cooper and other debris, including carburized metal, shall be removed by grinding or other mechanical methods approved by the INSPECTION AGENCY. Oxygen gouging of quenched and tempered steels or other high strength steels is not permitted.

For partial repairs, the cut-out portion shall be sufficiently deep and long to remove the defect. At the ends and sides of the cut, there shall be a gradual taper from the base of the cut to the surface of the weld metal. The width and profile of the cut shall provide adequate access for rewelding.

Special care shall be taken to remove weld defects located at the root in order to obtain an acceptable root gap.

Prior to starting the repair, the repair grooves shall be examined by dye penetrant method in accordance with ASME Code Section V, to ensure that all defects are removed.

Preheating (where required) and interpass temperatures shall be maintained during all weld repairs.

The repaired weld shall be subject to post weld heat treatment if required.

All repaired welds shall be re-inspected in accordance with this specification at the MANUFACTURER's expense. Moreover, additional radiographs of two welds previously performed by the same welder shall be required at the MANUFACTURER's expense. These additional two welds shall be selected by the INSPECTION AGENCY.

For materials other than austenitic stainless steels which are subject to impact tests at temperatures below 0°C, or for stainless steels used in corrosion environment, a second repair of the same area on a given weld shall not be permitted : should a defect be repaired on a previously repaired weld area, the whole weld shall be cut and rewelded after having restaured the base materials to their initial condition (i.e. remove heat affected zones, damaged surfaces, etc...).

For all other welds, only two repairs shall be permitted on the same area of a given weld : should a third repair become necessary, then the whole weld shall be cut as per above.

12 - WELDER IDENTIFICATION

In field welding, each qualified welder or welding operator shall have an identification symbol assigned to him and, unless specified otherwise, shall permanently mark each pressure weld with his identification symbol. If more than one welder welds a joint, each shall apply his symbol in a manner to indicate the part of the joint he welded.

For shop welds, the MANUFACTURER shall keep a record of the welder(s) employed on each joint.

13 - RECORDS OF INSPECTION

The MANUFACTURER shall keep a complete and accurate current record of all inspections and their results. These records shall be available for examination by the INSPECTION AGENCY at all reasonable times.